CPSC 292: Introduction to Exploratory Data Analysis

Lecture in which CLO is introduced | Assignments (C-level) | Projects (B-level) | Skill Checks (A-level)

Detailed Course Learning Objectives:

At the completion of this course, students should be able to:

1. Understand the basic structure and function of the *R* programming language.
   1.1. Understand the *R* Workspace.
      1.1.1. Understand how to use the command line. | L1.4 | A1.5 | P1 | SK1
      1.1.2. Understand how to use the help function of *R*. | L1.4 | A1.5 | P1 | SK1
      1.1.3. Understand the basic syntax of the *R* language. | L1.4 | A1.5 | P1 | SK1-SK3
      1.1.4. Execute inbuilt mathematical functions to perform calculations in *R*. | L1.4 | A1.5 | P1 | SK1-SK3
      1.1.5. Learn how to assign variables in *R*'s environment. | L1.4 | A1.5 | P1 | SK1
      1.1.6. Understand the basic syntax of functions in *R*. | L1.4 | A1.5 | P1 | SK1,SK3
      1.1.7. Open, edit and save a script in RStudio's editor. | L1.4 | A1.5 | P1-P2 | SK1-SK2
      1.1.8. Understand the concept of working directories. | L1.2, L1.4 | A1.1–1.3 | P1,P2,P4 | SK1,SK2
      1.1.9. Understand the data classes of *R*. | L1.5 | A1.6 | P1 | SK1
      1.1.10. Create vectors, arrays, matrices, lists, and data frames. | L1.5–1.9 | A1.6–1.12 | P1 | SK1
      1.1.11. Understand vectors and vectorized calculations. | L1.5–1.9 | A1.7 | P1 | SK1-SK3
      1.1.12. Learn how to index vectors, arrays, matrices, lists, and data frames. | L1.5–L1.9 | A1.7–1.12 | P1 | SK1-SK3
      1.1.13. List the special values in R. | L1.9 | A1.13 | P1 | SK1
      1.1.14. Create ordered and unordered factors. | L1.9 | A1.13 | P1 | SK1
      1.1.15. Learn to coerce data from one class into another. | L1.9 | A1.13 | P1 | SK1,SK2
   1.2. Understand the basic principles of computer programming.
      1.2.1. Understand the way computers execute commands. | L1.2, L1.3 | A1.1, A1.2, A1.4 | P1-P4 | SK1-SK3
      1.2.2. Create functions in *R*. | L3.1, L3.7, L3.8 | P2 | SK3
      1.2.3. Use functions to reduce repetitive procedures in a script. | L3.7, L3.8 | P2 | SK3
      1.2.4. Use functions to automate and standardize the production of a product (e.g. a graph, an analysis). | L3.7, L3.8 | P2,P4 | SK2-SK3
      1.2.5. Create a function that vectorizes a calculation. | L3.7, L3.8 | P2 | SK3
      1.2.6. Understand and successfully execute a `while` loop. | L3.5 | P2 | SK3
      1.2.7. Understand and successfully execute conditional `if/else` statements (vectorized and non-vectorized). | L3.4 | P2 | SK3

- 1.2.8. Understand and successfully execute `repeat` and `for` loops. | L3.5 | P2 | SK3
- 1.2.9. Use a loop to automate a calculation or procedure. | L3.6 | P2,P4 | SK3
- 1.2.10. Use a conditional statement to automate a calculation or procedure. | L3.6 | P2 | SK3

1.3. Understand the basic principles of software design.

- 1.3.1. Understand why and how code should be documented. | L3.1 | P2,P4 | SK3
- 1.3.2. Create documentation within code and outside of code in the form of a README file. | L3.10 | P2,P4 | SK3
- 1.3.3. Use whitespace effectively to make scripts more readable. | L1.10, L3.6, L3.10 | P2,P4 | SK3
- 1.3.4. Create and maintain R Project files in RStudio. | L3.2, L3.3, L3.10 | P3,P4 | –
- 1.3.5. Understand the basic principles of refactoring. | L3.6, L3.11 | P2,P4 | SK3
- 1.3.6. Execute code refactoring toward a specific goal (e.g. improving speed, improving readability). | L3.11 | P2,P4 | SK3

2. Understand and follow best practices in scientific computing.

2.1. Properly organize your work.

- 2.1.1. Understand how file systems are structured and organized. | L1.2, L3.1 | A1.2 |SK1, SK3 | P2
- 2.1.2. Understand how to navigate file systems using a GUI interface. | L1.2 | A1.2 |SK1, SK3 | P2
- 2.1.3. Use directories to organize course work. | L1.2 | A1.3 |SK1, SK3 | P2
- 2.1.4. Use standard organization to organize an R project. | L3.1 | SK3 | P2

2.2. Understand the importance of reproducibility in scientific data analysis.

- 2.2.1. Create reproducible scripts in *R*. | L1.4, L2.2, L3.1, L3.10 | P2,P4 | SK2,SK3
- 2.2.2. Include effective documentation in scripts and projects. | L2.2, L3.1, L3.10 | P2 | SK3,SK4
- 2.2.3. Understand the Open Science movement and the role of data repositories in research. | L3.1,L3.10 | P2,P4 | SK3
- 2.2.4. Create and use Notebooks and documents using RMarkdown. | L2.2, L3.10, L4.4, L4.5 | P1,P2,P4 | –

3. Independently perform basic data analysis and visualizations in a way that communicates ideas clearly.

3.1. Load, clean, and organize data in *R*. | L2.3 | SK2 | P1,P2,P4

- 3.1.1. Load text, CSV, Excel data files and built-in package data sets. | L2.3 | P1,P2,P4 | SK2
- 3.1.2. Clean, arrange, and transform data sets. | L2.3 | P1,P2,P4 | SK2
- 3.1.3. Learn to index specific values in data sets. | L2.3 | P1,P2,P4 | SK2
- 3.1.4. Create and use data frames from other data types. | L2.3 | P1,P2,P4 | SK2

3.2. Learn how to plot quickly using *R*'s base graphics. | L2.2 | P1-P4 | SK2
  - 3.2.1. Create bar graphs, line, scatter, box, and histogram plots. | L2.2 | P1 | SK2
  - 3.2.2. Learn when to implement each type of plot above. | L2.2 | P1 | SK2
  - 3.2.3. Create a plot legend. | L2.2 | P1 | SK2
  - 3.2.4. Alter the axes, title, and labels of a plot. | L2.2 | P1 | SK2
  - 3.2.5. Create text labels on a plot. | L2.2 | P1 | SK2

3.3. Learn the basics of `ggplot2`. | L2.6 | P1-P4 | SK2
  - 3.3.1. Understand the basic grammar of graphics used by `ggplot2`. | L2.6 | P1,P2,P4 | SK2
  - 3.3.2. Create line, scatter, box, and histogram plots with `ggplot2`. | L2.6 | P1,P2,P4 | SK2
  - 3.3.3. Make multi-panel plots. | L2.6 | P1,P2,P4 | SK2
  - 3.3.4. Create a plot legend. | L2.6 | P1,P2,P4 | SK2
  - 3.3.5. Alter the axes, title, and labels of a plot. | L2.6 | P1,P2,P4 | SK2
  - 3.3.6. Create text labels on a plot. | L2.6 | P1,P2,P4 | SK2

3.4. Execute qualitative and quantitative data analyses. | L2.5 | P1,P2,P4 | SK2,SK3
  - 3.4.1. Learn the the basic ways data are described. | L2.5 | P1-P4 | SK2,SK3
  - 3.4.2. Learn the **`head`**, **`tail`**, and **`summary`** commands. | L2.5 | P1-P4 | SK2
  - 3.4.3. Calculate the mean, median, quartiles, range, and standard deviation of a data set. | L2.5 | P1-P4 | SK2

3.5. Think and work independently with code. | L1.10, L2.4, L2.5, L2.6, L3.9, L.3.10, L3.11 | P1-P4 | SK1-SK3
  - 3.5.1. Learn basic skills in debugging and troubleshooting error messages. | L1.10, L2.4, L3.8 | A1.14 | P1-P4 | SK1-SK3
  - 3.5.2. Search for effective solutions and tools using online resources. | L2.4, L3.8 | P1-P4 | SK1-SK3
  - 3.5.3. Learn to accept and provide constructive feedback to peers. | L2.10, L2.11, L2.12 | SK2.1, SK2.2 | P1, P2

3.6. Understand the basic principles of data visualization and communication.
  - 3.6.1. Understand the basic way our brain processes light stimuli into sight. | L2.7 | P1-P4 | SK2
  - 3.6.2. Understand the role of contrast in low-level visual processing. | L2.7, 2.9 | P1-P4 | SK2
  - 3.6.3. Understand the organization of the 'what' and 'where' visual processing systems. | L2.7, 2.8 | P1-P4 | SK2.2
  - 3.6.4. Understand how acuity is focused centrally in vision and drops off as you move into the periphery. | L2.7, L2.8 | P1-P4 | SK2
  - 3.6.5. Understand the importance of attention and accessibility to communicating ideas. | L2.8 | P1-P4 | SK2

3.6.6. Understand the importance of ethical presentation of data through visualization. | L2.8 | P1 | SK2

3.6.7. Understand the principle of the Curse of Knowledge and how it impedes communicating ideas. | L2.9 | P1-P4 | SK2

3.7. Learn the most effective strategies for written and oral presentation of scientific ideas.

3.7.1. Discuss the reasons for communicating scientific results to a variety of audiences. | L4.1 | P3-P4 | –

3.7.2. Practice reading and understanding scientific papers and grant proposals. | L4.2 | P4 | –

3.7.3. Practice watching and understanding oral presentations (slideshow and poster). | L4.3 | P3 | –